Make training work

VLC = ∫∫∫ Vision Learning Control $dV_d L_d C$

Practical Training of Deep Networks

Normalisation, stabilisation, and diagnosis

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Training is a coupled dynamical system

Initialisation recap: Xavier, He, and PyTorch defaults

Normalisation layers control activation scale

Batch normalisation

BatchNorm during training and evaluation

Batch size and data dependence

Layer normalisation

Initialisation and normalisation complement each other

Gradient clipping limits unstable updates

Learning-rate schedules and warm-up

Weight decay connects optimisation and regularisation

Training curves give diagnostic evidence

Inspecting activations and gradients

Dead and saturated units

A systematic failure-diagnosis workflow